ITAI 1371 ? Module 10

INDIVIDUAL REFLECTION JOURNAL

Alexander DeBusk

Unsupervised Learning

L10 – Alexander DeBusk – Reflection

In this module, I learned about unsupervised learning and how it can be used to find patterns in data without needing labeled answers. I worked with K-Means clustering to group data points based on similarities. I also learned how the number of clusters affects the results and how the Elbow Method can help determine a reasonable number of clusters by comparing inertia values.

Another topic I learned about was Principal Component Analysis (PCA). PCA reduces the number of features in a dataset while keeping as much of the important information as possible. I used PCA on the Iris dataset to reduce its four original features to two principal components. This made the data easier to visualize while still preserving most of its variance.

One challenge was understanding what the different parameters in the code controlled and how they affected the results. Working through the code helped me understand how clustering and dimensionality reduction work in practice. I also became more familiar with using Python libraries such as Scikit-learn, Matplotlib, and Seaborn to analyze and visualize data.

ITAI 1371 ? Module 10

INDIVIDUAL REFLECTION JOURNAL

Shareefah Adeola

REFLECTION JOURNAL – **Module 10: Unsupervised Learning**

ITAI 1371

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In this module, I learned how unsupervised learning identifies patterns in data without predefined labels. How unsupervised learning differs from supervised learning and how it can be used to discover patterns in data when there is no target variable provided. I practiced the two important techniques: K-Means clustering and Principal Component Analysis (PCA).

Using the Elbow Method, I compared inertia for k values from 1 to 10 and selected four clusters. I then applied K-Means to group the synthetic data and visualized the clusters and their centers. This made the relationship between the data points and their centroids easier to understand

I also used PCA to reduce the Iris dataset from four features to two principal components. The two components retain approximately 97.77% of the total variance, showing that the data could be represented in two dimensions while preserving most of its variation as using the Iris dataset, the first principal component explained approximately 92.46% of the variance and the second explained approximately 5.31%. Together, the two components explained approximately 97.77% of the total variance. This result helped me understand that dimensionality reduction can make data easier to visualize while preserving most of its variation.

One challenge was placing code in the correct notebook sections and correcting an indentation error. I also learned that cluster labels must be included in the plotting command to display groups in different colors. Working through these issues improved my confidence in using Python and interpreting machine-learning results.

Overall, this lab strengthened my understanding of clustering, dimensionality reduction, data visualization, and the importance of checking code and interpreting outputs.

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INDIVIDUAL REFLECTION JOURNAL

Yilin Leng

Unsupervised Learning

Reflection

Through this lab, I gained a better understanding of unsupervised learning by working with K Means clustering and Principal Component Analysis. I learned that unsupervised learning could find patterns in unlabeled data and group similar data together. This can be useful in real businesses, such as understanding customer preferences and finding unusual patterns. In the K Means experiment, I learned that using more clusters does not always mean better results. By applying the Elbow Method, I found that K = 4 was a good choice for the Iris dataset. This helped me understand the importance of balancing model performance and complexity. I also learned how PCA can reduce the number of features while keeping most of the important information. In this experiment, the first two principal components retained 97.77% of the total variance, making the data easier to visualize and analyze. However, these algorithms also have some limitations. PCA may not work well with complex nonlinear data, while K Means can be affected by outliers and the starting positions of cluster centers. In the future, I would like to learn more about SNE, UMAP, and DBSCAN to understand how they work with different types of data. Overall, this lab helped me understand how unsupervised learning can find useful patterns in data and why choosing the right algorithm is important for different projects.